

DysCalc ML and Data Science Expert Evaluation

Repository under review:	https://github.com/DysCalc/DysCalc-ML-Development.git
Thesis Study Title:	DysCalc: An AI-Assisted Learning Platform for Screening and Targeted Learning for Students with Potential Dyscalculia in the Philippine Educational Context
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Evaluation Purpose:	Independent expert validation of the repository's data cleaning, exploratory data analysis, machine learning implementation, model evaluation, synthetic data augmentation, and reproducibility.
Evaluator Profile:	Professional Statistician/Data Scientist

Evaluation Context and Instructions

DysCalc is a thesis-level proof-of-concept system that uses machine learning to support early screening of learners who may show signs of numeracy-related difficulty. The machine learning pipeline uses the FUNA-DB dataset to classify learners into two screening categories: Typical and At-Risk. The model output is intended to support educator review and instructional planning. It is not intended to serve as a clinical diagnosis of dyscalculia.

This validation form is designed to assess the methodological soundness of the DysCalc machine learning pipeline. The evaluation focuses on the appropriateness of the data preprocessing steps, feature preparation, label construction, algorithm selection, GAN-based synthetic data generation, statistical validation of synthetic samples, model training, evaluation metrics, and interpretation of results.

As an expert validator, you are asked to review whether the pipeline is technically appropriate for a thesis-level proof-of-concept screening system. Please evaluate the methods based on their data science, statistical, and machine learning validity. You may provide comments, required revisions, or recommendations where necessary.

For each criterion, please indicate one of the following:

[A] Approved — The criterion is fully met and methodologically sound.
[R] Needs Revision — The criterion is partially met but requires correction, clarification, or stronger justification.
[N/A] Not Applicable — The criterion does not apply to this specific implementation.

Please provide concise feedback in the **Expert Notes** column, especially for any item marked as **Needs Revision**. And at the end of each section, please provide a section-level rating:

- Excellent	- Needs Minor Revision
- Satisfactory	- Needs Major Revision

A. Data Preprocessing and Feature Preparation

Item	Required Condition	Status (A / R)	Comments
A1	The preprocessing steps are appropriate for preparing the FUNA-DB dataset for machine learning.		
A2	The handling of missing, invalid, or sentinel values is methodologically reasonable.		
A3	The selected raw features, such as NC, DM, NS, ADD, SUB, and CA, are appropriate for representing numeracy-related performance.		
A4	The derived features are logically connected to numeracy constructs and are suitable for model input.		
A5	The preprocessing procedure avoids data leakage between training, validation, and test sets.		
A1	The preprocessing steps are appropriate for preparing the FUNA-DB dataset for machine learning.		
A2	The handling of missing, invalid, or sentinel values is methodologically reasonable.		
A3	The selected raw features, such as NC, DM, NS, ADD, SUB, and CA, are appropriate for representing numeracy-related performance.		
Section Rating: ☐ Excellent ☐ Satisfactory ☐ Needs Minor Revision ☐ Needs Major Revision			
Evaluator Notes:			

B. Data Preprocessing and Feature Preparation

Item	Required Condition	Status (A / R)	Comments
B1	The construction of the At-Risk and Typical labels is clearly explained.		
B2	The use of RMAT or an external arithmetic fluency reference for label derivation is appropriate for screening purposes.		
B3	The label construction is separated from the predictive features to avoid leakage.		
B4	The classification target is appropriately framed as a screening indicator rather than a clinical diagnosis.		
Section Rating: ☐ Excellent ☐ Satisfactory ☐ Needs Minor Revision ☐ Needs Major Revision			
Evaluator Notes:			

C. Algorithm Selection and Model Appropriateness

Item	Required Condition	Status (A / R)	Comments
C1	The chosen machine learning algorithm is appropriate for the size and structure of the dataset.		
C2	The chosen algorithm is suitable for a binary classification task		

	involving Typical and At-Risk learners.		
C3	The use of an interpretable model is appropriate because the system is intended to support educator decision-making.		
C4	The model implementation and training procedure are methodologically sound.		
C5	The selected model is appropriate for proof-of-concept screening rather than automated diagnosis.		
Section Rating: ☐ Excellent ☐ Satisfactory ☐ Needs Minor Revision ☐ Needs Major Revision			
Evaluator Notes:			

D. GAN-Based Synthetic Data Generation

Item	Required Condition	Status (A / R)	Comments
D1	The use of synthetic data generation is justified by the class imbalance in the training set.		
D2	The use of GAN-based tabular synthesizers, such as CopulaGAN and CTGAN, is appropriate for generating additional minority-class training samples.		
D3	The GAN models are trained only on the real minority-class training samples and do not use validation or test data.		
D4	The number of synthetic At-Risk samples generated is appropriate for balancing the training set.		
D5	The post-processing of synthetic data, including feature constraints and recomputation of derived features, is methodologically reasonable.		
D6	The handling of missingness patterns in synthetic data is appropriate and avoids producing unrealistic complete records.		
Section Rating: ☐ Excellent ☐ Satisfactory ☐ Needs Minor Revision ☐ Needs Major Revision			
Evaluator Notes:			

E. Statistical Validation of Synthetic Data

Item	Required Condition	Status (A / R)	Comments
E1	The comparison between real and synthetic At-Risk samples is appropriate for validating synthetic data quality.		
E2	The Kolmogorov-Smirnov test is appropriate for assessing distributional similarity between real and synthetic feature values.		
E3	Jensen-Shannon Divergence is appropriate for measuring distributional distance between real and synthetic samples.		
E4	Summary-statistic comparisons, including mean, standard deviation, median, skewness, and kurtosis, are appropriate for evaluating central tendency, spread, and distribution shape.		
E5	The validation criteria are sufficient to support the use of synthetic samples for training augmentation.		
E6	The conclusion that synthetic At-Risk samples are statistically similar to real At-Risk samples is supported by the validation		

	results.		
Section Rating:	☐ Excellent ☐ Satisfactory ☐ Needs Minor Revision ☐ Needs Major Revision		
Evaluator Notes:			

F. Model Training and Evaluation Design			
Item	Required Condition	Status (A / R)	Comments
F1	The dataset split into training, validation, and test sets is appropriate.		
F2	The validation and test sets being kept fully real is appropriate for fair evaluation.		
F3	The comparison between TRTR and TRSTR is appropriate for evaluating the effect of synthetic augmentation.		
F4	The model evaluation design avoids synthetic data leakage into validation or test results.		
F5	The selected training strategy is justified by validation and held-out test performance.		
Section Rating:	☐ Excellent ☐ Satisfactory ☐ Needs Minor Revision ☐ Needs Major Revision		
Evaluator Notes:			

G. Performance Metrics and Results Interpretation			
Item	Required Condition	Status (A / R)	Comments
G1	The selected metrics are appropriate for an imbalanced binary screening task.		
G2	Recall is appropriately prioritized because false negatives may delay support for At-Risk learners.		
G3	F2-score is appropriate because it gives greater weight to recall than precision.		
G4	Precision and accuracy are interpreted appropriately as trade-offs rather than ignored.		
G5	The comparison between TRTR and TRSTR is interpreted correctly.		
G6	The results support the claim that synthetic augmentation improved sensitivity to At-Risk learners.		
G7	The study avoids overclaiming the model as a diagnostic tool.		
Section Rating:	☐ Excellent ☐ Satisfactory ☐ Needs Minor Revision ☐ Needs Major Revision		
Evaluator Notes:			

H. Overall ML Pipeline Validity

Item	Required Condition	Status (A / R)	Comments
H1	The overall ML pipeline is methodologically sound for a thesis-level proof-of-concept study.		
H2	The preprocessing, feature engineering, model training, synthetic augmentation, and evaluation steps are logically connected.		
H3	The pipeline is appropriate for early numeracy screening support.		
H4	The limitations of the ML pipeline are acknowledged appropriately.		
H5	The ML pipeline is suitable for professional validation at this stage of development.		
H1	The overall ML pipeline is methodologically sound for a thesis-level proof-of-concept study.		
H2	The preprocessing, feature engineering, model training, synthetic augmentation, and evaluation steps are logically connected.		

Section Rating: ☐ Excellent ☐ Satisfactory ☐ Needs Minor Revision ☐ Needs Major Revision

Evaluator Notes:

FINAL VALIDATION

Based on the review of the above criteria and minimum passing gates, please select the final validation status for this repository:

	Validation Status	Description
☐	Validated	The ML pipeline is methodologically sound and supports the thesis proof-of-concept claims.
☐	Validated with Minor Revisions	The ML pipeline is sound but requires minor clarification, documentation, or small corrections.
☐	Conditionally Validated with Major Revisions	The ML pipeline requires substantial methodological corrections before it can be considered valid.
☐	Not Yet Validated	The ML pipeline is not yet methodologically acceptable in its current form.

Required Revisions or Recommendations

Please list any required changes or recommendations.

REVIEWER INFORMATION

Reviewer Name	
Affiliation / Institution	
Position / Title	
Area of Expertise	
Email Address	
Date of Review	

Signature